ADVANCED MATERIALS

Supporting Information

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Data-Driven Additive Discovery with HOMO-Descriptor Enables Durable Aqueous Zinc Batteries via Interfacial Kinetics Engineering

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1. Experimental Section

1.1. Materials

Zinc sulfate heptahydrate ($ZnSO_4·7H_2O$, ≥99.0%) and sulfuric acid (H_2SO_4, 95.0–98.0%) was purchased from Sinopharm Chemical Reagent Co., Ltd. Pyridine (Pyr, ≥99.0%), 4-Picoline (Pyr-CH_3, ≥98.0%), 4-(Methylamino)pyridine (Pyr-$NHCH_3$, ≥99.0%), 4-Dimethylaminopyridine (DMAP, ≥99.0%), 4-Pyridinecarbonitrile (Pyr-CN, ≥98.0%), 4-Hydroxypyridine (Pyr-OH, ≥97.0%), 4-Nitropyridine (Pyr-NO_2, ≥95.0%), 4-Methoxypyridine (Pyr-OCH_3, ≥98.0%) and Vanadium pentoxide (V_2O_5, ≥99.0%) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Sodium chloride (NaCl, ≥99.0%) and N-methyl pyrrolidone (NMP, AR) were purchased from Shanghai Macklin Biochemical Technology Co., Ltd. Polyvinylidene fluoride (PVDF, Arkema), stainless steel welded mesh (400 mesh), Ketjen Black, graphite felt (1 mm in thickness) and glass fiber membrane were used as received. Battery cases and other consumables were purchased from Guangdong Canrd New Energy Technology Co., Ltd.

1.2. Preparation of Electrolytes and Cathode Materials

The 2 M $ZnSO_4$ (ZSO) electrolyte was prepared by dissolving an appropriate amount of $ZnSO_4·7H_2O$ in deionized water. Pyridine derivative-modified electrolytes were subsequently prepared by adding Pyr, Pyr-CH_3, Pyr-$NHCH_3$, DMAP, Pyr-CN, Pyr-OH, Pyr-NO_2, or Pyr-OCH_3 to the 2 M ZSO electrolyte, maintaining a constant additive concentration of 0.1 M. The corresponding electrolytes are marked as ZSO-Pyr, ZSO-Pyr-CH_3, ZSO-Pyr-$NHCH_3$, ZSO-DMAP, ZSO-Pyr-CN, ZSO-Pyr-OH, ZSO-Pyr-NO_2 and ZSO-Pyr-OCH_3 electrolyte, respectively. To verify the optimal concentration of DMAP, 2 M ZSO electrolytes with 0.05 M and 0.2 M DMAP were prepared and marked as ZSO-0.05M DMAP and ZSO-0.2M DMAP, respectively. All pH values of the pyridine derivatives-modified electrolytes were adjusted by a sulfuric acid solution. The preparation of $NaV_3O_8·1.5H_2O$ (NaVO) was based on the previous research.[1] Briefly, 1 g V_2O_5 powder was dissolved in 15 mL 2 M NaCl solution followed by stirring at 500 rpm for 96 hours in a water bath at 30 ℃. After that, NaVO powder was obtained by washing with deionized water, followed by filtration and freeze-drying. The NaVO electrode was fabricated by mixing the active materials (NaVO), Ketjen Black and binder (PVDF) with a mass ratio of 7:2:1 in NMP and blade the slurry onto stainless steel

welded mesh (diameter: 12 mm) and dried in a vacuum oven at 80 °C overnight. The active material loading was controlled to approximately 1 mg cm^{-2}.

1.3. Materials Characterizations

The microscopic morphologies of Zn foils and NaVO powder were investigated by field emission scanning electron microscopy (SEM, Tescan Mira4), confocal laser scanning microscope (CLSM, KEYENCE VK-X1000) and atomic force microscope (AFM, Bruker Dimension Icon). X-ray diffraction (XRD) patterns were obtained by X-ray diffractometer (Rigaku Mini Flex 600 diffractometer) with Cu-Kα radiation and a scan rate of 10° min^{-1}. X-ray photoelectron spectroscopy (XPS) spectra were characterized by a XPS spectrometer (Thermo Fisher-VG Scientific) with Al Kα radiation. Raman spectra of electrolytes were obtained via Raman spectrometer (LabRAM HR800). In situ Raman spectra measurements were performed by acquiring signals from the electrolyte near the electrodes in a custom-made transparent symmetric cell with the Raman spectrometer, while the cell was charged at 2mA cm^{-2} by an electrochemical workstation (CHI660E, China). The optical photographs of the Zn anode during plating process were obtained by coupling an optical microscope (CEWEI LW750LJT) and the custom-made transparent symmetric cell, and a current density of 5mA cm^{-2} was employed. Operando pH variations of electrolytes were recorded by PHS-25 meter (INESA Scientific) with an automatic sampling system (JENCO, 6173R) under a constant current density of 1mA cm^{-2}.

1.4. Electrochemical measurements

Electrochemical performance was evaluated using CR2025-type coin cells assembled in ambient atmosphere. Zn foil (thickness: 100 μm, 99.99%) was cleaned by ethanol and punched into 15 mm diameter discs to serve as Zn electrodes. Each cell contained 100 μL of electrolyte. Galvanostatic charge/discharge cycling was performed at 25℃ using a NEWARE battery testing system (CT-4008T5V50mA-164, CT-4008T-5V6A-S1-F). Zn|Zn symmetric cells were assembled using two Zn electrodes separated by a glass fiber separator (Whatman GF/D, 15 mm diameter). Zn|Cu asymmetric cells, employing a Zn anode and a Cu foil cathode (20 μm thickness, 99.99% purity), were used to assess average Coulombic efficiency (CE) with a charging cutoff voltage of 0.6 V. The NaVO|Zn full

cell was assembled by Zn electrode as anode and NaVO electrode as cathode with a cycling voltage range of 0.3-1.5V. Cyclic voltammetry (CV), Tafel, Chronoamperograms (CA), linear scan voltammetry (LSV) and electrochemical impedance spectroscopy (EIS) were tested on an electrochemical workstation (CHI 604E, China). All CV tests employed a scan rate of 0.1 mV s^{-1} within specific voltage ranges: Zn|Cu (-0.2 V to 0.6 V), and NaVO|Zn (0.3 V to 1.5 V). CA tests were performed using Zn|Zn cells with a fixed overpotential of -200 mV. LSV curves were carried out on Zn|Zn symmetric cells at a scan rate of 1mV s^{-1}. EIS spectra of Zn|Zn cells were acquired under the frequency range of 10 mHz to 100 kHz. The differential electrochemical mass spectrometry (Shanghai Linglu Instrument Equipment Co., Ltd) was utilized to record the H_2 generation at the Zn anode. The in-situ electrochemical digital holography (EDH, Prosilica GT1910) was performed to record the variation of ion concentration at the electrode/electrolyte interface and the Zn|Zn cells were assembled in a transparent optical cell and charged by a CHI660E. For the assembly of the pouch cells, the cathode was prepared by coating a graphite felt substrate (7 × 8 cm^2 for 0.3 Ah level, 8 × 10 cm^2 for 1.2 Ah level) with active material slurry (~14 mg/cm^2), while zinc foil (thickness: 80 μm, Squre: 7.5 × 8.5 cm^2 for 0.3 Ah level and 8.5 × 10.5 cm^2 for 1.2 Ah level) served as the anode. Ti foil was used as the cathode current collectors. The 0.3 Ah level pouch cell employed a single zinc foil anode paired with a single NaVO cathode and contained 9 mL of electrolyte. Similarly, the 1.2 Ah pouch cell utilized a stacked assembly comprising three zinc foil anodes and three NaVO cathodes, with an electrolyte volume of 39 mL. For the pouch cell module powering the robotic dog, each individual cell unit featured a stacked configuration of two zinc foil anodes (8.5 × 10.5 cm^2) and two NaVO cathodes (8 × 10 cm^2), containing 20 mL of electrolyte. Four such cell units were connected in series to form the battery module. The charging process followed a constant current-constant voltage (CC-CV) protocol: constant current charging until the voltage reached 1.5 V, followed by constant voltage charging to maintain the voltage while reducing the current to one-third of the initial rate to compensate for capacity. The exchange current density (i_0) was determined via LSV in a three-electrode configuration, utilizing two Zn electrodes (working and counter electrodes) and an Ag/AgCl reference electrode, and the measurement was performed with a scan rate of 0.1 mV s^{-1} from -6 to 6 mV vs. open circuit potential (OCP). The calculation of Exchange current density (i_0) followed the equation:

$$i_0 = \frac{IRT}{F\eta} \quad (1)$$

where I, R, T, F and η represent the applied current density (A cm^{-2}), the gas constant (J mol^{-1} K^{-1}), the absolute temperature (K), the Faraday constant (96500C mol^{-1}) and the total overpotential (V), respectively.

2. Computational methods

2.1. Multi-physics simulation

The finite element simulation was performed utilizing COMSOL Multiphysics 6.2 software. Specifically, a Zn anode surface model measuring 100 × 100 μm² with an average roughness of 17.5 μm was constructed, implementing this roughness within COMSOL via Gaussian and uniform random distribution functions.[2]. The multi-physics simulation model was further established using an ultra-fine mesh with a maximum mesh size of 0.020 μm. The dynamic evolution of electric fields and Zn^{2+} concentration on the Zn surface during electrodeposition was simulated by COMSOL's deformation mesh functionality. The model incorporated the Butler-Volmer equation, Faraday's law of electrolysis, and the Nernst-Einstein relationship to govern the electrochemical reactions at the Zn surface. Critically, the MUMPS solver was used to handle the model-solving process, with the assumption of no parasitic reactions, corresponding to 100% electrodeposition efficiency.

2.2. Quantum chemistry calculations

The density functional theory (DFT) method was performed using the Gaussian16 program.[3] with Becke's three-parameter hybrid method using the Lee-Yang-Parr correlation functional (B3LYP)[4] at 6-311+G(d,p) level. All optimized structures were confirmed as potential minima, with no frequency modes with imaginary eigenvalues, through frequency analyses following geometry optimizations. Based on the optimized structures, molecular dipole moments, polarizability were analyzed. The highest occupied molecular orbital (HOMO) energy level, the lowest unoccupied molecular orbital (LUMO) energy level and atomic charges were analyzed using the Natural Bond Orbitals (NBO) theory[5]. The molecular orbitals and electrostatic potential (ESP) calculation results were visualized by using the Multiwfn 3.8 program and the visual molecular dynamics (VMD) program.[6]

2.3. Ab initio calculations

All first principle calculation (Ab initio) calculations were conducted by using the Vienna Ab initio simulation package (VASP).[7] The generalized gradient approximation (GGA) functionals were considered by Perdew-Burke-Ernzerh as well as the projector augmented wave method (PAW-PBE) to describe exchange-correlation effects between electrons.[8] To calculate the adsorption energy, the supercells with two-layer Zn slabs were built to represent the adsorbed surface for molecules and the bottom layer was fixed to maintain the bulk property. A 20.0 Å vacuum layer was built along the Z-axis of supercells to limit interactions between periodic images. The energy cutoff for the planewave basis set was 400 eV, which was accompanied with the convergence criterion for the electronic self-consistent field (SCF) and loop set to 1×10^{-7} eV/atom. The geometry optimization of all structures applied a 7 × 7 × 7 Gamma k-point grid and conducted until the residual forces were below 0.01 eV $Å^{-1}$. The calculation of adsorption energy (E_{ads}) followed the equation:

$$E_{ads} = E_{Zn+mol} - E_{Zn} - E_{mol} \quad (2)$$

where the E_{Zn+mol}, the E_{Zn}, and the E_{mol} represent the total energy of adsorbates on the Zn surface, the energy of the Zn surface, and the energy of the isolated adsorbates, respectively.

2.4. Machine Learning

The well-established symbolic ensemble algorithm Adaptive Boosting (AdaBoost) was employed to construct the machine learning model. AdaBoost iteratively combines multiple weak learners into a strong learner by dynamically reweighting training samples. Samples misclassified by a learner receive increased weight in subsequent iterations, focusing the model on increasingly difficult cases and progressively improving performance. This work employed decision stumps as base weak learners with n_estimators = 50 and learning_rate = 1.0. The model was initially trained on a dataset comprising molecular descriptors (D, P, q^+, and E_{max}) for 24 organic additives. Its predictive accuracy was then evaluated using an independent test set containing descriptors for 6 organic additives. After the evaluation and adjustment of the model, we established the final predictive model through training on a dataset containing above 30 organic additives.

Pearson correlation coefficient (PCC) and SHapley Additive exPlanation (SHAP) were utilized to analyze the correlation between input descriptors and HOMO energy. The Pearson correlation coefficient (PCC) ranges between -1 and 1. A positive value indicates a positive correlation between

the two features, while a negative value signifies a negative correlation. The magnitude of the absolute value reflects the strength of the correlation. For the SHAP analysis presented in **Figure 1g**, features are ordered along the vertical axis according to their importance in influencing the model output. The lower horizontal axis represents the SHAP value, which depicts the distribution of each feature's impact on the model output. Each point in the plot corresponds to an individual sample, with color indicating the relative magnitude of the feature value. Taking the top-ranked feature P as an example (top row), high P values (represented by red dots) are associated with increased model predictions, whereas conversely, low P values (represented by blue dots) are associated with decreased model predictions.

Table S1. Training data for ML prediction. Molecules No. 1-30 as the training set and No. 31-37 as the prediction set. Molecules No. 20-33 as the verifying set for HOMO-CPC relationship.

No.	Mole-cule	HO-MO	LU-MO	D	P	q^+	q^-	E_{max}	E_{min}	CPC	Ref
		eV	eV	Debye	a.u.			Kcal/mol		Ah	
1	DM	-7.05	-0.32	2.27	99.74	0.485	-0.771	47.75	-35.47	3.50	[9]
2	Scm	-7.17	-0.08	1.79	93.41	0.461	-0.741	42.54	-39.84	1.40	[10]
3	DU	-6.27	-0.19	4.25	89.55	0.832	-0.685	19.18	-46.73	2.00	[11]
4	Glu	-7.09	-0.82	1.34	82.73	0.808	-0.848	54.17	-33.51	1.00	[12]
5	I2959	-6.59	-1.62	5.75	164.36	0.555	-0.738	48.53	-58.32	1.00	[13]
6	SA	-8.11	-0.69	2.40	61.10	0.807	-0.702	52.31	-32.39	3.75	[14]
7	VER	-6.57	-1.86	4.90	123.40	0.416	-0.568	26.55	-39.85	2.00	[15]
8	Phe	-6.78	-0.70	1.26	119.95	0.797	-0.835	50.89	-30.14	0.75	[16]
9	FA	-7.47	-0.19	4.05	25.31	0.523	-0.811	48.59	-42.37	0.50	[17]
10	AA	-7.17	-0.27	3.98	36.88	0.663	-0.813	46.84	-44.62	0.63	
11	TFA	-8.33	-1.08	4.10	37.61	1.000	-0.789	53.25	-34.06	3.00	
12	BT	-6.70	-0.70	2.99	51.59	0.577	-0.672	19.68	-38.00	1.50	[18]
13	GBL	-7.66	-0.23	4.91	50.67	0.807	-0.564	23.85	-42.17	3.13	[19]
14	Pyr	-7.22	-1.13	2.37	61.03	0.213	-0.453	21.43	-37.66	1.38	[20]
15	BA	-6.93	-0.17	4.36	55.67	0.683	-0.649	37.37	-45.97	2.50	[21]
16	NMS	-7.93	-0.49	5.45	59.47	2.194	-0.919	44.14	-39.55	3.25	[22]
17	PG	-7.53	-0.25	2.82	47.58	0.470	-0.765	45.98	-38.10	0.50	[23]
18	G2	-7.18	0.19	1.42	89.27	0.185	-0.610	12.16	-29.97	0.73	[24]
19	G3	-7.19	0.19	0	118.78	0.185	-0.609	12.30	-29.67	1.90	
20	G4	-7.19	0.18	1.42	148.42	0.185	-0.609	12.34	-29.61	2.50	
21	HEI	-6.26	-0.40	4.96	75.69	0.466	-0.734	49.71	-46.95	3.13	[25]
22	Cit	-7.04	-0.49	3.37	109.47	0.797	-0.842	50.81	-44.95	3.00	[26]
23	ETH	-7.63	-0.05	1.76	30.91	0.457	-0.744	40.36	-36.35	2.50	[27]
24	EG	-7.79	-0.16	0	34.78	0.461	-0.742	42.78	-31.99	2.00	[28]
25	Ace	-7.05	-0.76	3.15	40.16	0.57	-0.673	20.65	-38.07	2.50	[29]
26	BD	-7.02	-2.74	0	52.54	0.495	-0.678	25.27	-28.65	3.25	[30]
27	FUC	-7.32	-1.99	4.21	66.76	0.804	-0.643	60.10	-36.90	2.25	[31]
28	SL	-7.89	-0.36	5.68	69.97	2.004	-0.920	24.69	-40.94	2.00	[32]
29	TEP	-7.95	-0.10	3.99	108.26	2.519	-1.053	15.57	-49.22	1.75	[33]
30	TMU	-6.46	-0.10	3.49	82.17	0.833	-0.657	16.72	-45.09	3.75	[34]
31	Pyr-CH_3	-7.13	-0.98	2.89	74.39	0.220	-0.597	18.89	-39.40	2.68	
32	Pyr-CN	-7.94	-2.48	2.01	78.07	0.282	-0.430	27.54	-33.52	0.08	
33	Pyr-NO_2	-8.07	-3.40	2.16	78.01	0.485	-0.424	26.34	-27.84	0.25	
34	DMAP	-5.97	-0.39	4.84	99.20	0.219	-0.495	26.01	-44.82	4.25	

35	Pyr-NHCH$_3$	-6.15	-0.49	4.64	86.15	0.380	-0.610	42.64	-44.26	0.43	
36	Pyr-OCH$_3$	-7.04	-0.73	3.40	79.53	0.352	-0.533	25.19	-40.15	0.88	
37	Pyr-OH	-7.27	-0.84	2.74	66.53	0.473	-0.664	57.69	-38.56	0.10	

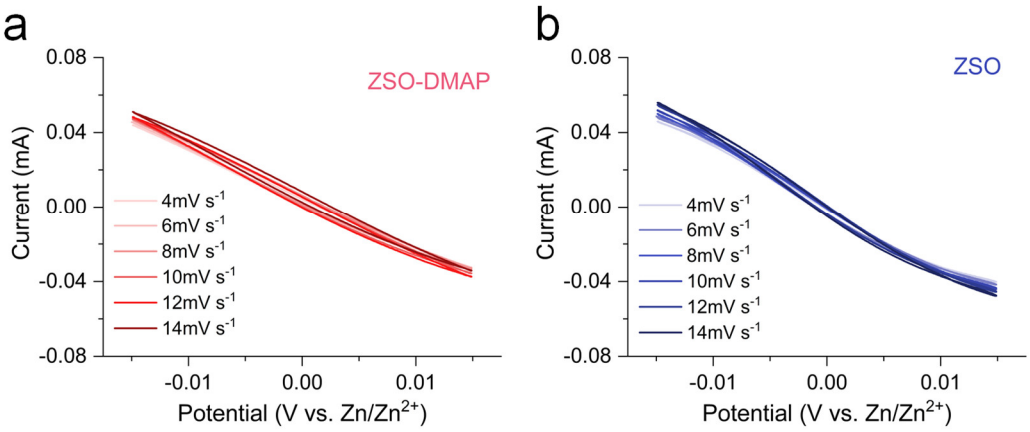


Figure S1. CV curves of Zn|Zn cells utilizing **a)** ZSO-DMAP and **b)** ZSO electrolytes in a voltage range of -15 mV to 15 mV under different scanning rates.

The double-layer capacitance (c) was determined from the linear relationship between the capacitive current (i_c) and the scan rate (v), using the slope of the i_c versus v plot. This relationship is expressed by the equation:

$$c=i_c/v \tag{3}$$

where i_c represents the capacitive current obtained from CV scans. Here, we utilize $i_c=(i_{0+}-i_{0-})/2$, meaning the half value of the current difference during the forward scan and negative scan at 0 V.

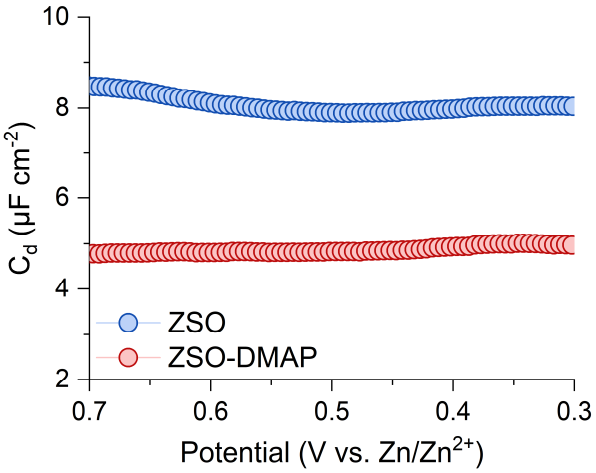


Figure S2. Differential capacitance curves of Zn|Zn cells with ZSO and ZSO-DMAP electrolytes.

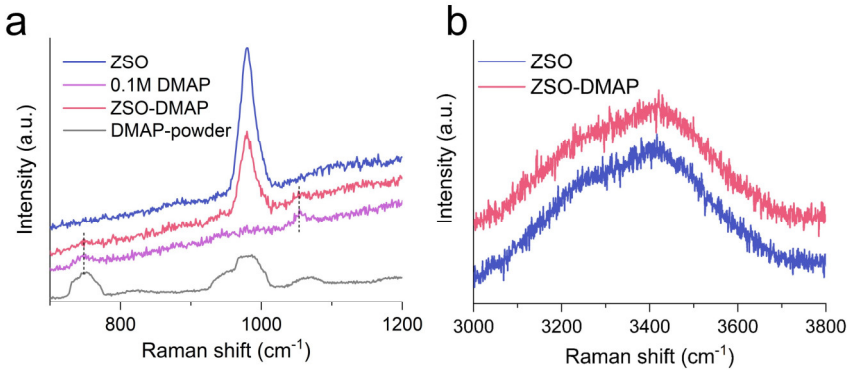


Figure S3. Raman spectra of ZSO and ZSO-DMAP electrolyte with amplified **a)** SO_4^{2-} and **b)** O-H peaks

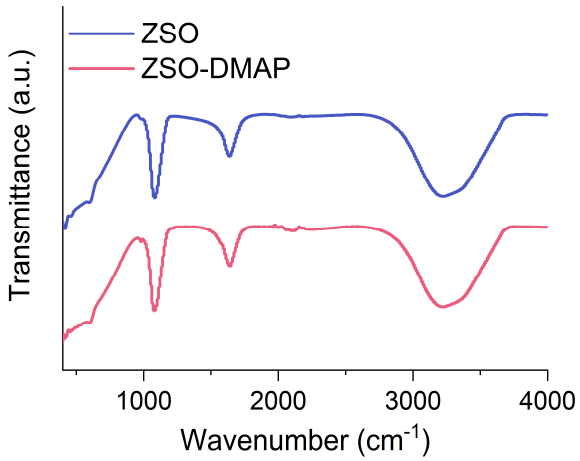


Figure S4. FTIR spectrum of ZSO and ZSO-DMAP electrolytes

Raman spectral analysis reveals significant peaks variation in 0.1 M DMAP aqueous solution compared to pure DMAP, indicative of interaction between DMAP and H_2O. Subsequent addition of $ZnSO_4$ induces no substantial shift in DMAP-associated vibrational modes. These observations collectively demonstrate minimal involvement of DMAP in zinc ion solvation structures at this low concentration. Furthermore, this low concentration proves insufficient to substantially perturb the hydrogen-bonding network of bulk water, as corroborated by **Figure S3b** and **S4**. Consequently, DMAP's influence on zinc deposition behavior is not principally mediated through bulk-phase modifications. Rather, the dominant mechanism involves interfacial regulation at the electrode/electrolyte boundary, where trace DMAP concentrations effectively modulate Zn deposition kinetics.

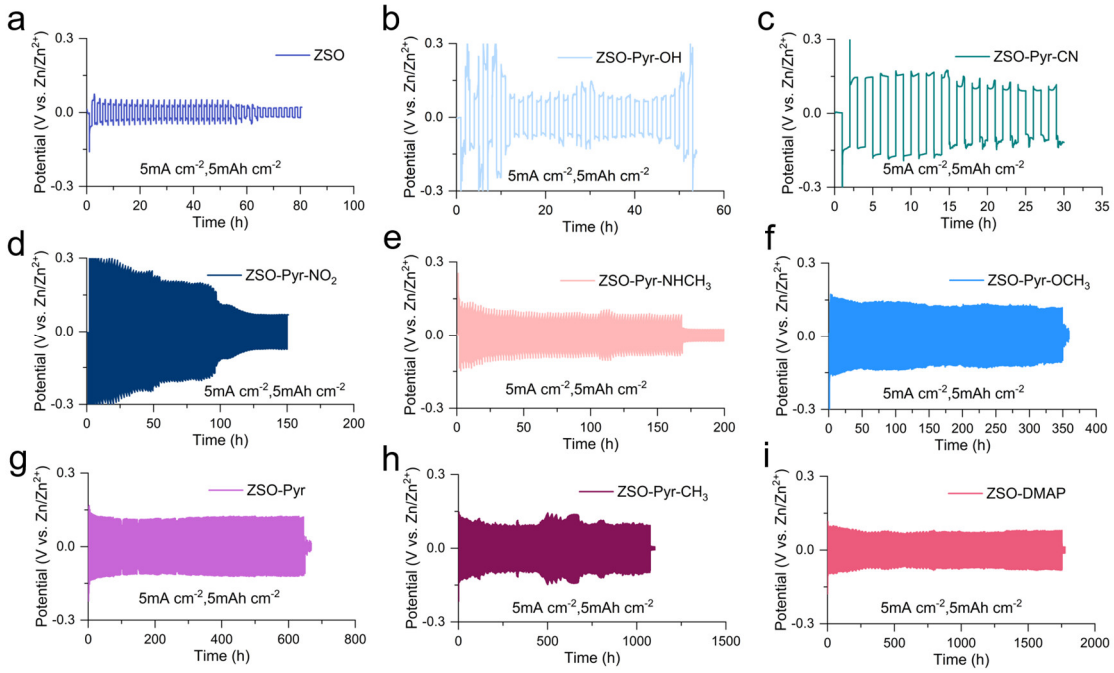


Figure S5. Long-term cycling performance of Zn|Zn symmetric cells at 5mA cm^{-2}, 5mAh cm^{-2} in different pyridine derivative-modified electrolytes.

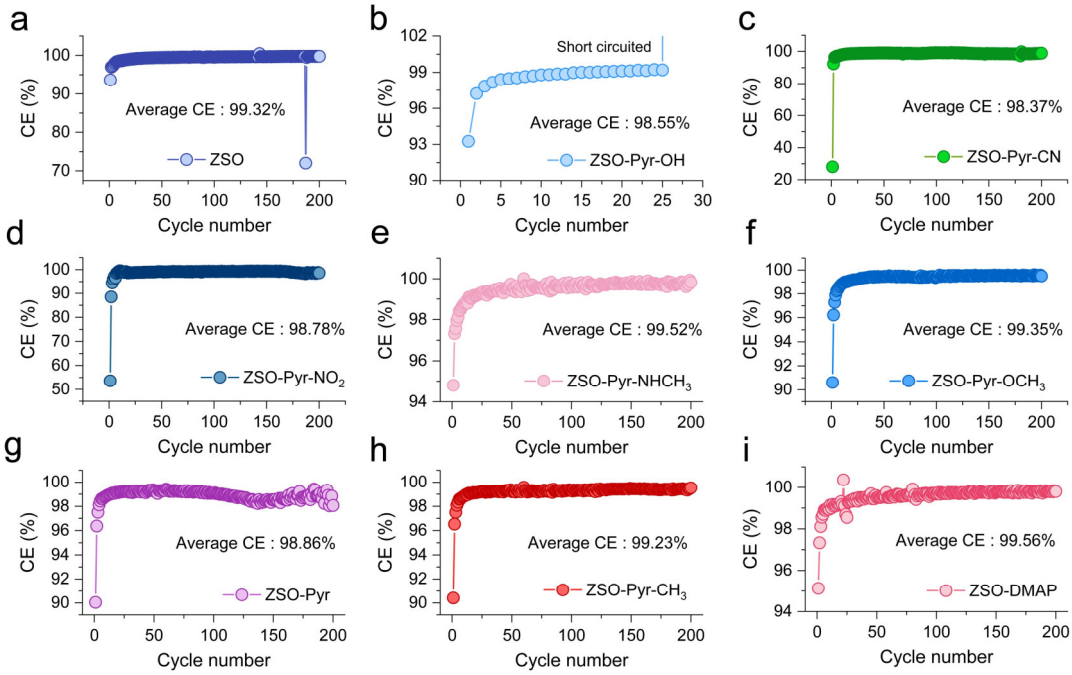


Figure S6. CEs of Zn|Cu cells operated at 1mA cm^{-2}, 1mAh cm^{-2} in different pyridine derivative-modified electrolytes.

Symmetric cell cycling tests demonstrated that cell with DMAP-modified electrolyte exhibited stable cycling for 1600 hours under stringent conditions (5 mA cm^{-2}, 5 mAh cm^{-2}), significantly surpassing other pyridine derivatives-modified electrolytes. Furthermore, Coulombic efficiency (CE) measurements revealed that the DMAP-modified electrolyte maintained an average CE of 99.56% over 200 cycles. These results collectively substantiate DMAP as the optimal additive for achieving highly reversible and stable zinc anodes.

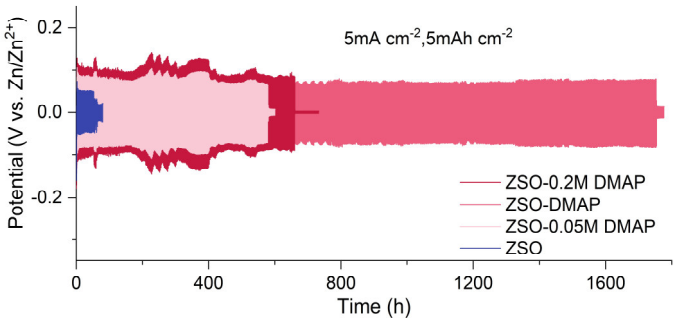


Figure S7. Long-term cycling performance of Zn|Zn symmetric cells at 5mA cm^{-2}, 5mAh cm^{-2} in ZSO electrolytes with different DMAP concentrations.

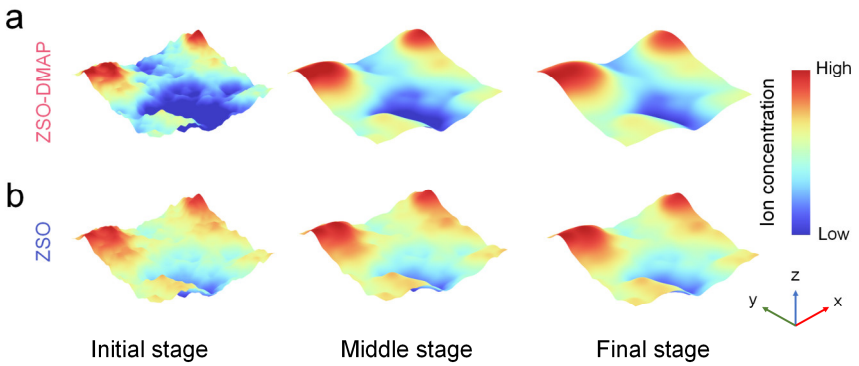


Figure S8. Evolution of the ionic concentration on Zn anode surface with the electrolyte of **a)** ZSO-DMAP and **b)** ZSO at initial, middle, and final stages respectively.

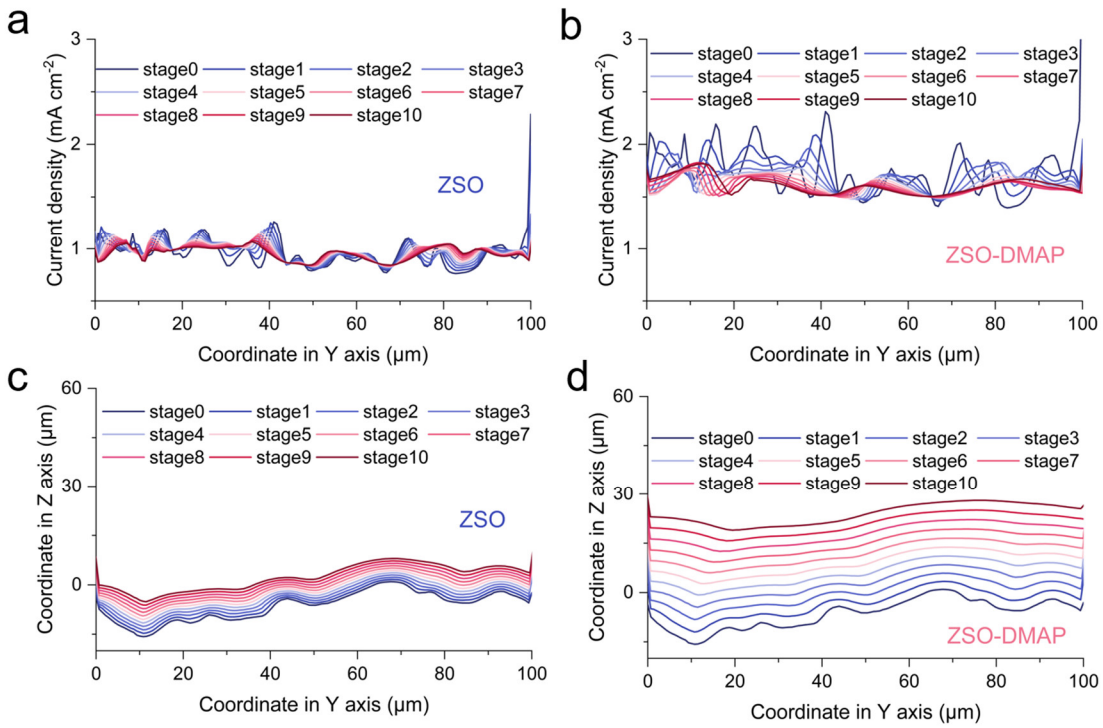


Figure S9. Variation of current density on Zn surface along Y-axis at X = 50 μm in **a)** ZSO and **b)** ZSO-DMAP electrolytes. Corresponding height profile of Zn surface during the plating stage in **c)** ZSO and **d)** ZSO-DMAP electrolytes.

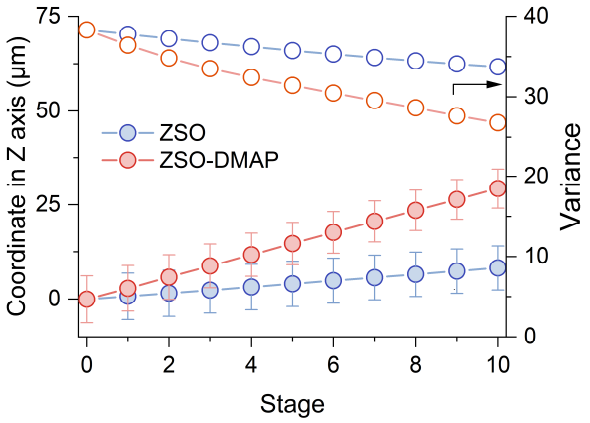


Figure S10. Change in the Coordinate in Z axis and uniformity of the Zn anode during the complete electrodeposition process with different electrolytes

The statistical analysis of the coordinate in Z axis reflects the variation of height profile of the Zn anode surface. The height variance of Zn anode surface in ZSO-DMAP experiences a more distinct reduction compared to ZSO, indicating a more uniform Zn electrodeposition endowed by DMAP.

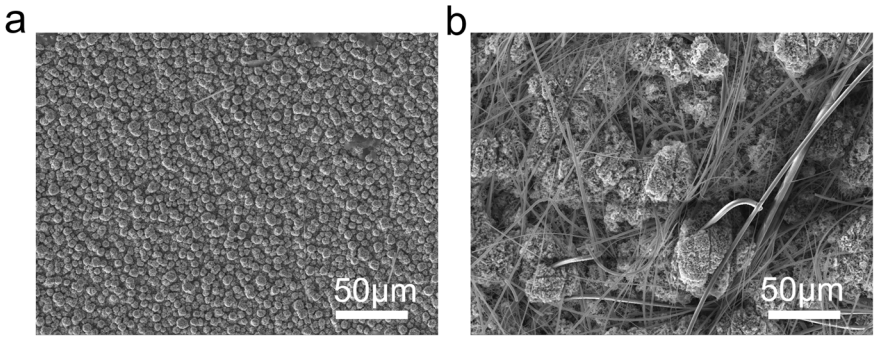


Figure S11. SEM images of Zn deposits (2 mAh cm^{-2}) on Cu substrate in **a)** ZSO-DMAP and **b)** ZSO electrolytes.

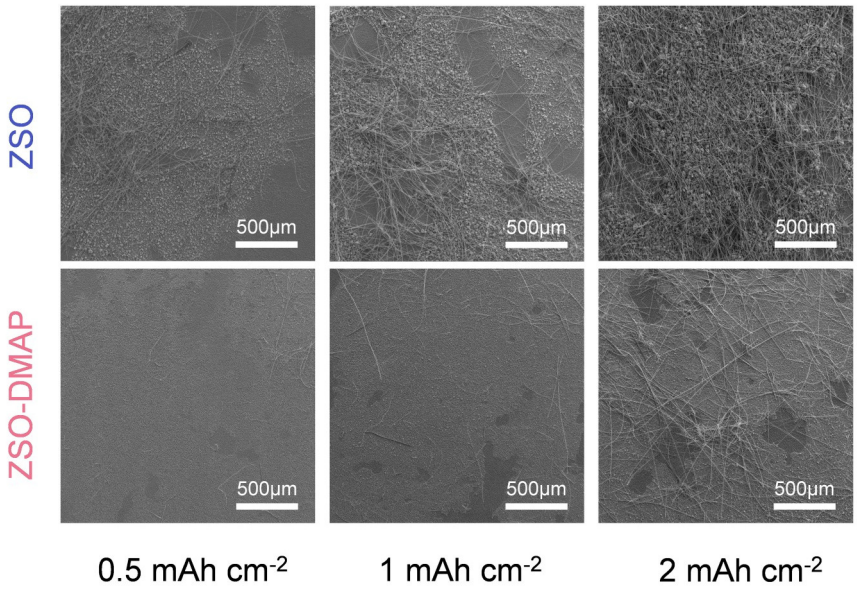


Figure S12. Low magnification SEM images of Zn electrodeposition on Cu substrate in ZSO and ZSO-DMAP.

The Zn electrodeposition morphology studies were conducted in Zn|Cu coin cells assembled with glass fiber separators. In the unmodified ZSO electrolyte, the Zn nucleation and growth on the copper substrate are highly heterogeneous, often resulting in localized dendritic structures. As shown in **Figure S12**, these dendrites readily interlock with the fibrous matrix of the separator, forming a mechanically inseparable composite. This entanglement makes it difficult to fully remove the glass fiber residues during cell disassembly, which explains the persistent fibrillar morphology observed in the SEM images of ZSO samples (**Figure 4f** and **Figure S11**). In contrast, when DMAP is introduced into the electrolyte (ZSO-DMAP), its interfacial modulation effect leads to more uniform and denser Zn electrodeposition. This suppresses dendritic overgrowth and significantly reduces physical interaction with the separator fibers. As a result, the electrodeposited Zn layer can be isolated more cleanly from the glass fiber separator, and the corresponding SEM images of ZSO-DMAP samples exhibit minimal to no fiber contamination.

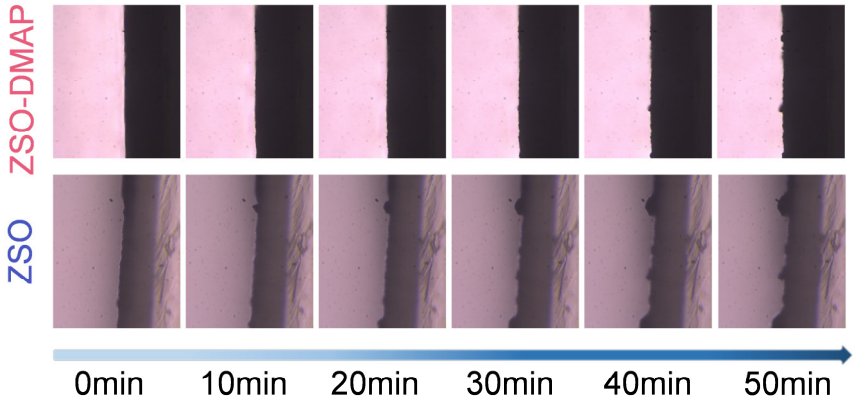


Figure S13. Visualized Zn plating process in ZSO and ZSO-DMAP electrolytes

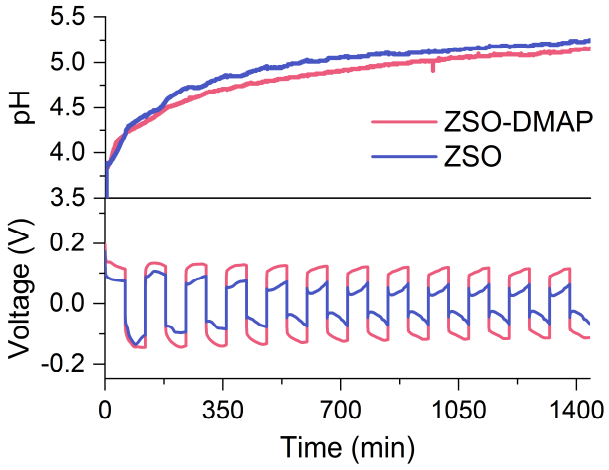


Figure S14. In-situ pH measurement of electrolytes during plating/stripping.

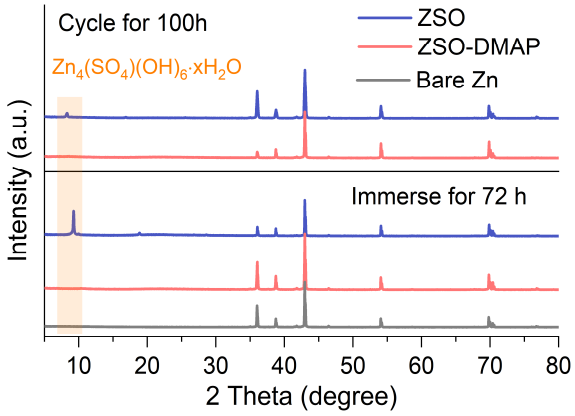


Figure S15. XRD patterns of Zn anode after immersing for 72 hours and cycling for 100 h at 1 mA cm^{-2}, 1 mAh cm^{-2} in ZSO and ZSO-DMAP electrolytes.

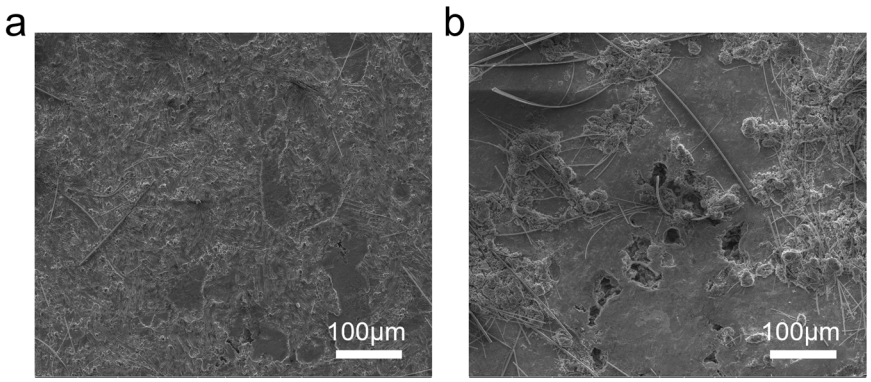


Figure S16. SEM images of Zn electrodes after 50 cycles in **a)** ZSO-DMAP and **b)** ZSO electrolytes.

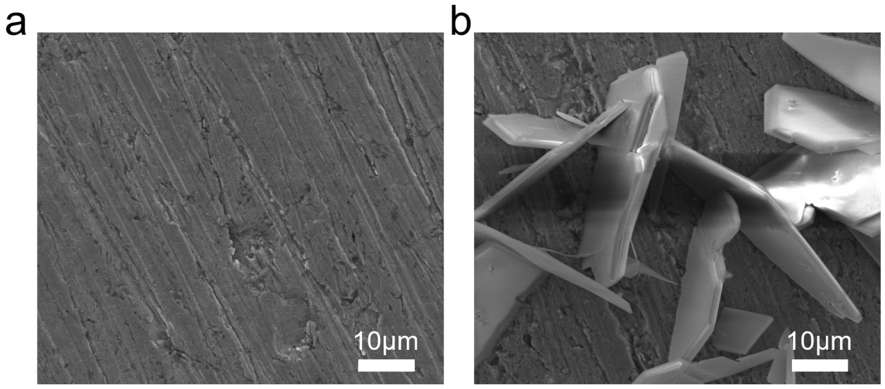


Figure S17. SEM images of Zn electrodes after being immersed in **a)** ZSO-DMAP and **b)** ZSO for 72 h.

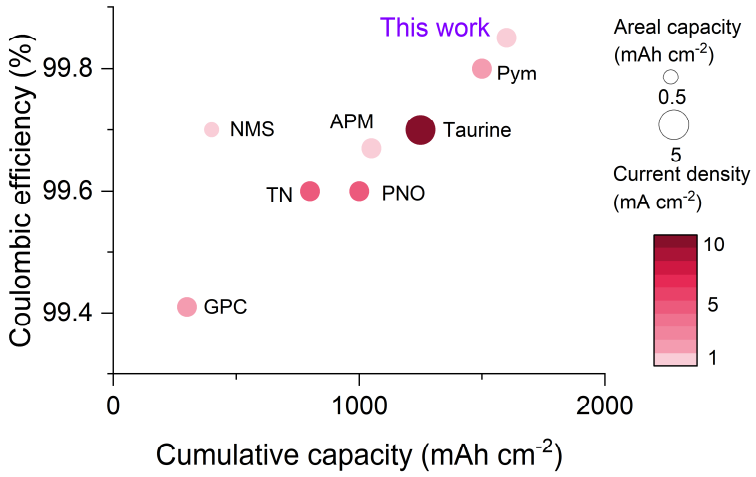


Figure S18. The Zn|Cu cell performance comparison of recent demonstrations with our work.[35]

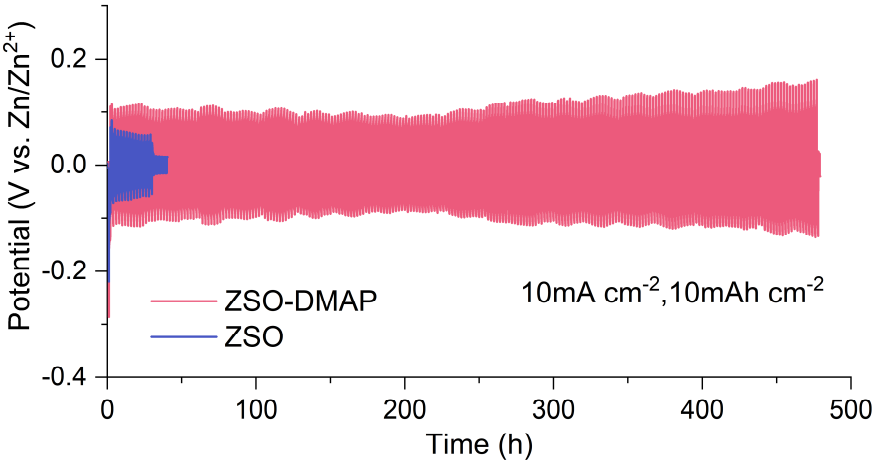


Figure S19. Long-term cycling performance of Zn|Zn symmetric cells at 10 mA cm^{-2}, 10 mAh cm^{-2}.

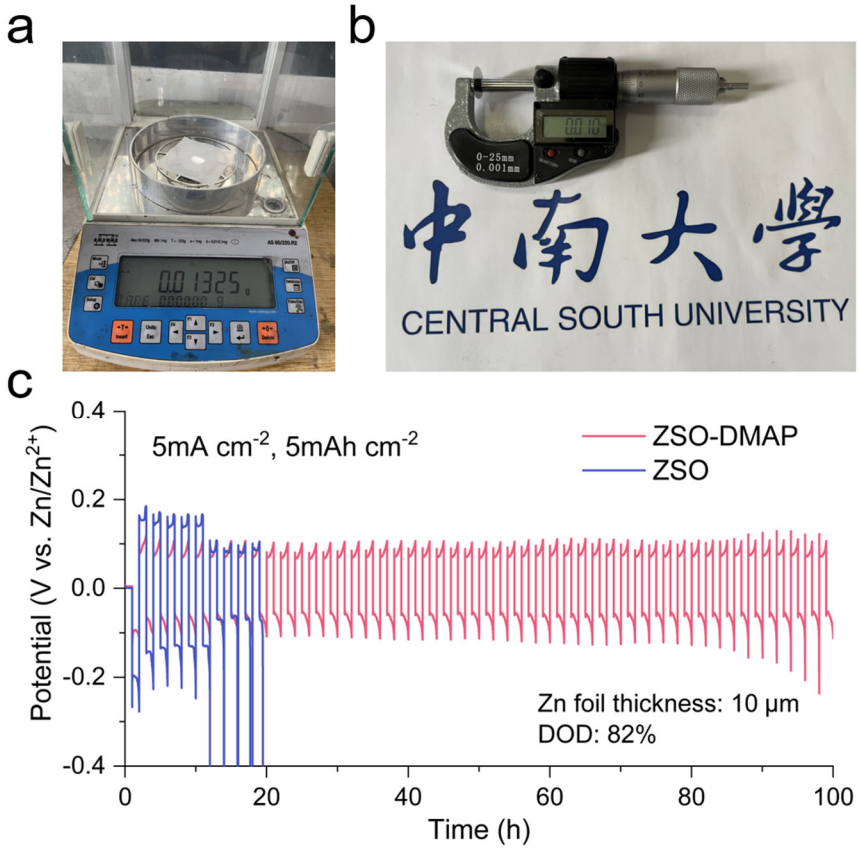


Figure S20. a) Weight and **b)** thickness measurements of the thin Zn electrode. **c)** Cycling stability of Zn|Zn symmetric cells with ZSO and ZSO-DMAP electrolytes at 5 mA cm^{-2}, 5 mAh cm^{-2}.

Under high current density and high areal capacity conditions (10 mA cm^{-2}, 10 mAh cm^{-2}), Zn|Zn symmetric cell employing ZSO-DMAP electrolyte maintained stable cycling for over 400 hours,

dramatically exceeding the performance of cell with ZSO electrolyte (40 hours). Furthermore, under stringent conditions at 82% depth of discharge (DOD), cell utilizing ZSO-DMAP electrolyte sustained stable operation for 100 hours. In contrast, cell employing ZSO electrolyte exhibited large polarization after merely 15 hours.

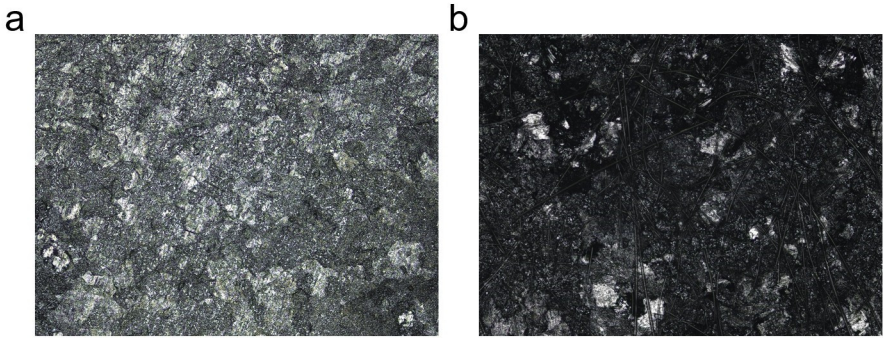


Figure S21. Unprocessed CLSM topographic image of a) ZSO-DMAP and b) ZSO corresponding to the results of the CLSM in **Figures 5f-h**.

Regarding the CLSM measurements presented in **Figures 5f–h**, we have carefully examined the unprocessed surface topography data (**Figure S21**) to assess potential interference from glass fibers. In the ZSO samples, some fiber residues are indeed present, either lying across the surface or embedded within the Zn layer. However, their influence on the calculated surface roughness is minor. The primary contributors to surface roughness are dendritic protrusions and porous features formed during heterogeneous Zn electrodeposition. The glass fibers contribute only marginally and do not affect the statistical comparison between ZSO and ZSO-DMAP systems.

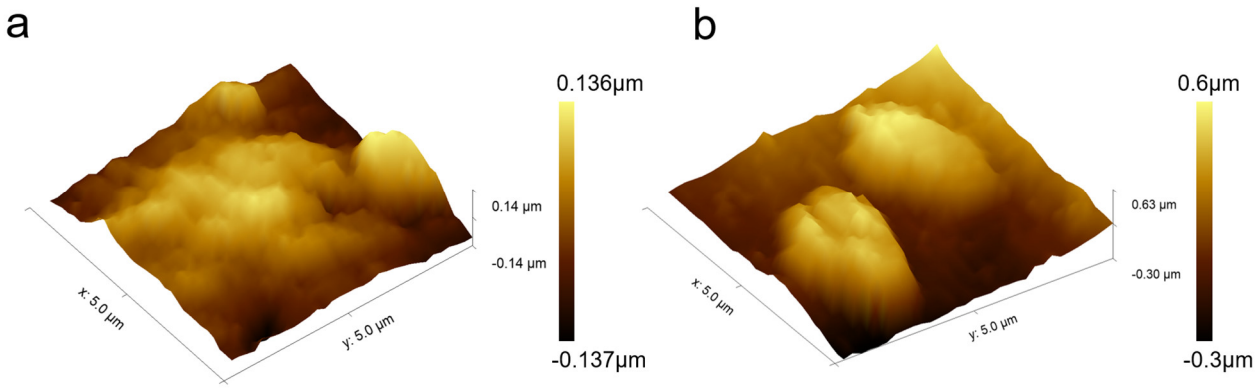


Figure S22. AFM images of Zn electrodes after 50 cycles in **a)** ZSO-DMAP and **b)** ZSO electrolytes.

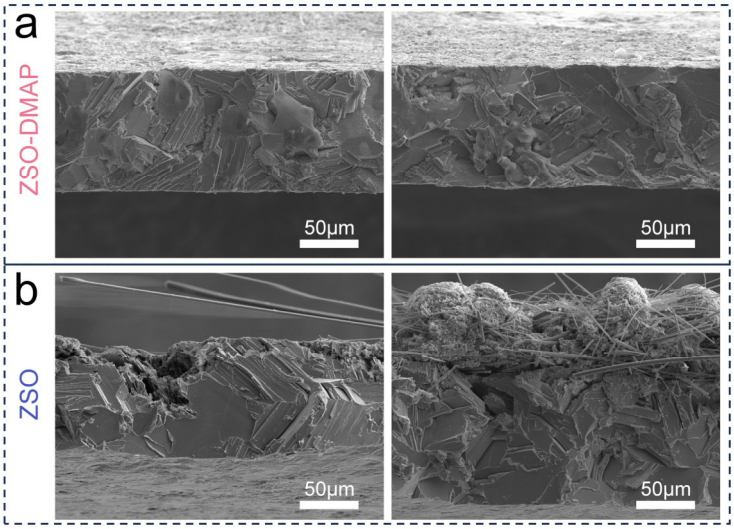


Figure S23. Cross-sectional SEM images of Zn electrodes after 50 cycles in a) ZSO-DMAP and b) ZSO electrolytes.

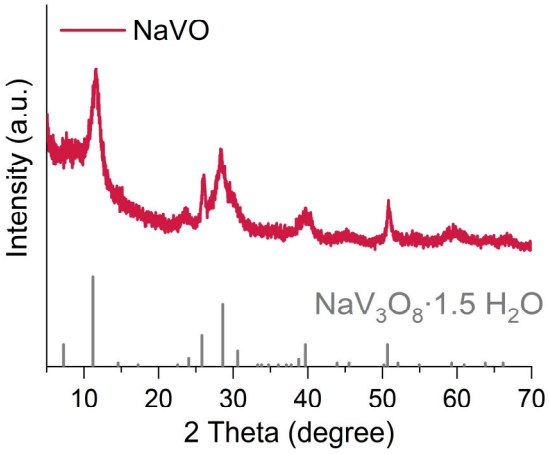


Figure S24. XRD pattern of as-prepared $NaV_3O_8{\cdot}1.5H_2O$

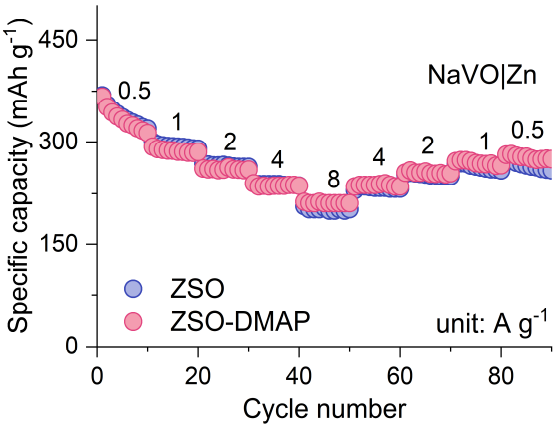


Figure S25. Rate performance of NaVO|Zn cells at different current densities from 0.5 to 8 A g^{-1} in ZSO and ZSO-DMAP electrolytes.

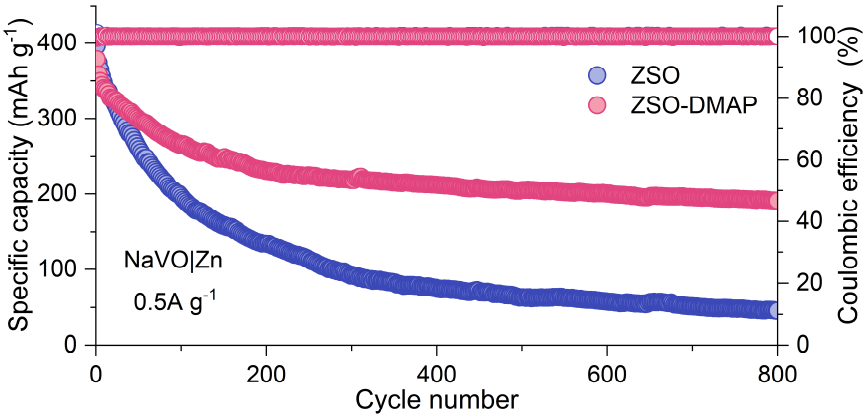


Figure S26. Cycling performances of NaVO|Zn full cells at 0.5A g^{-1}.

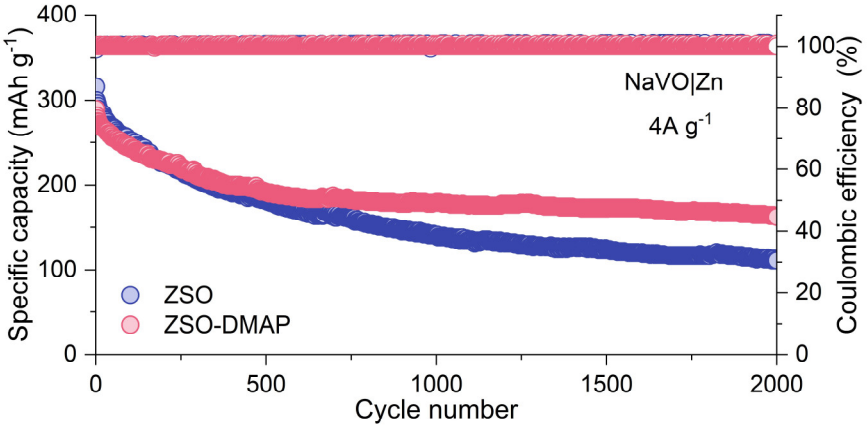


Figure S27. Cycling performances of NaVO|Zn full cells at 4A g^{-1}.

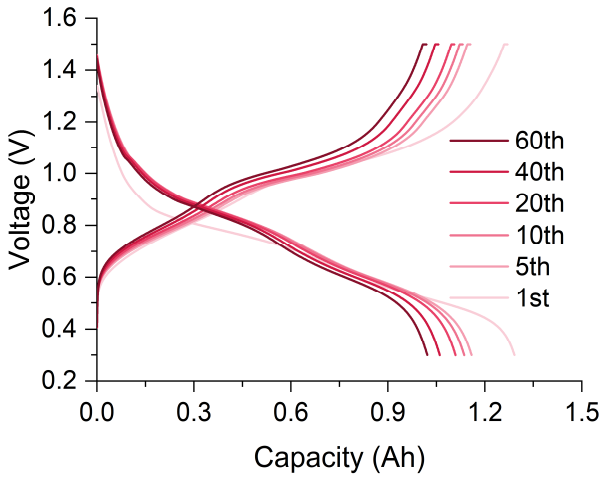


Figure S28. Discharge/charge curves of the 1.2 Ah level pouch cell with ZSO-DMAP electrolyte.

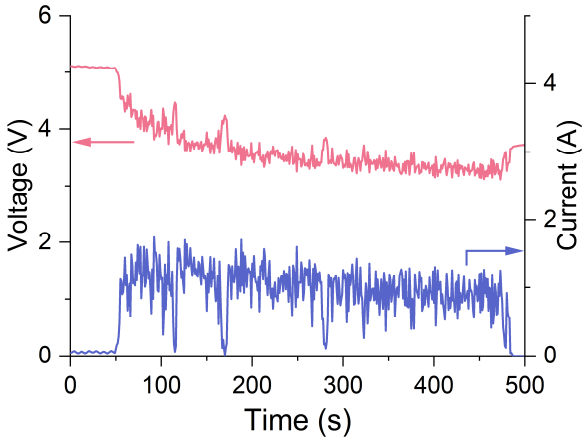


Figure S29. Voltage and current curves of the pouch cell to the robotic dog corresponding to the Video S1.

Table S2. The pouch full cell performance comparison of recent demonstrations with our work.

Additive	Cathode	Total capacity (mAh)	Cathode mass loading (mg cm^{-2})	Areal capacity (mAh cm^{-2})	Cycle number	Ref.
DMAP (This work)	$NaV_3O_8 \cdot 1.5H_2O$	319.6	14.27	5.71	250	/
pyrrolidone carboxylate sodium (PCA-Na)	α-MnO_2	370	30	3.7	80	[36]
sodium	MnO_2	140.28	33.4	5.85	200	[37]

benzenesulfonate (SBS)						
disodium naphtha-lenedisulfonate (NADS)	MnO_2	190	25	2.97	100	[38]
3-dimethyl(meth-acryloyloxyethyl) ammonium propane sulfonate (DM)	$NH_4V_4O_{10}$	37	20.9	2.31	50	[39]
YCl_3	$NH_4V_4O_{10}$	37.3	12.93	1.87	180	[40]
1-butyl-3-methylimidazolium cation ($BMIm^+$ ion)	$NH_4V_4O_{10}$	15	/	1.25	240	[41]
Sodium 3-mercapto-1-propanesulfonate (MPS)	$V_2O_5·H_2O$	31.7	7.7	1.28	250	[42]
sodium 3,3′-dithiodipropane sulfonate (SPS)	$V_2O_5·H_2O$	20.7	7	1.84	200	[43]
1,2-dimethoxyethane (DME)	$V_2O_5·H_2O$	62.4	9.33	2.08	100	[44]
1-carboxymethyl-3-methylimidazolium chloride (CMIM)	PANI	68	25.48	2.67	250	[45]
cucurbit[6]uril (CB[6])	PANI	14.7	13.5	1.6	340	[46]
Tetraphenylporphyri n tetrasulfonic acid (TPPS)	$NaV_3O_8·1.5H_2O$	350	18	5.75	80	[47]

KSeCN	V_2O_5	76.5	25	4.37	200	[48]

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